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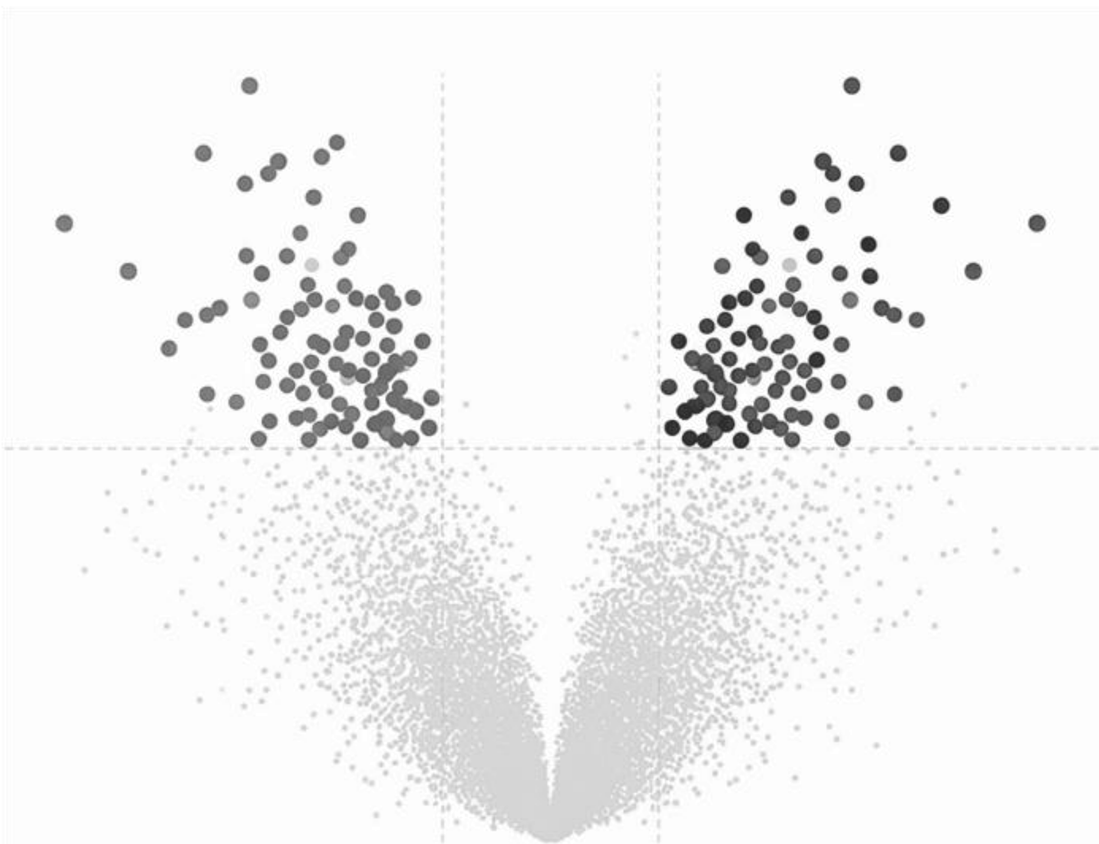
Bioinformatic analysis of host immune response

An omic approach using RNASeq in 195 patients



BRIEFLY DESCRIPTION

Bioinformatic analysis of a transcriptomic dataset comprising 195 patients. The study focused on characterizing the differential host immune response to SARS-CoV-2 versus bacterial pneumonia, enabling the identification of potential therapeutic targets through advanced RNA-Seq pipelines.

Work developed as part of my technical portfolio in Biostatistics and Omics Data Science.

Dataset: GSE161731 (Gene Expression Omnibus)　　March 2026
Methodology: bulk-RNASeq (Bioconductor)

RNASeq-Differential-Expression of SARS-CoV-2

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1. Introduction

This study, published in 2021 following the SARS-CoV-2 pandemic (McClain et al., 2021), explores the activation of biological processes and molecular markers in peripheral blood samples from infected patients. These findings are further compared against other respiratory conditions, including bacterial pneumonia, influenza virus infection, and healthy controls.

The analytical approach employed is mRNA transcriptomics using an universal library preparation. Sequencing was performed on the Illumina® NovaSeq 6000 platform. Stringent quality control filters were applied, including a minimum read length of 100 base pairs and a depth of at least 50 million reads per patient sample.

Collectively, this research provides an exhaustive understanding of the biological—primarily immunological—response associated with each respiratory pathology, enabling a detailed comparative analysis of their shared and distinct molecular signatures.

2. Objectives

- **Data Processing & Replication**: Using the raw data deposited in GEO (GSE161731), this project performs a complete end-to-end processing workflow. The analysis is conducted starting from raw counts, replicating the original investigators' rigorous approach to ensure data integrity.

- **Structured Transcriptomic Pipeline**: The project follows a systematic and ordered pipeline for analyzing transcriptomic data. This workflow transitions from raw data preprocessing and normalization to a final functional analysis, identifying the dysregulated biological pathways and molecular signatures associated with each respiratory pathology.

3. Methodology

The transcriptomic workflow was implemented in **R (v4.x)** using the **Bioconductor** framework. Information of packages and versions is available in *Session_info.txt* file. Raw sequencing counts and clinical metadata were retrieved from the **NCBI Gene Expression Omnibus (GEO)** under accession number **GSE161731** and integrated into a *SummarizedExperiment* object for downstream analysis. To ensure accurate feature attribution, genomic coordinates were mapped to the transcripts, followed by a rigorous data-wrangling process to filter out non-informative variables and standardize gene nomenclature.

To account for potential library composition biases across the 195 samples, we applied the **Trimmed Mean of M-values (TMM)** normalization method via the edgeR package. This "trimming" step was essential to stabilize variance and allow for robust inter-sample comparisons. Exploratory Data Analysis (EDA) was then performed using **Principal Component Analysis (PCA)** and **unsupervised hierarchical clustering**. For the latter, **Ward's Method (Ward.D2)** was utilized to minimize within-cluster variance, providing a high-sensitivity framework for the detection and subsequent removal of technical or clinical **outliers**.

Following quality control, **Differential Expression Analysis (DEA)** was conducted using the **DESeq2** package, which employs empirical Bayes shrinkage to estimate dispersion and fold-change. Finally, to characterize the biological significance of the results, we performed **Functional Enrichment Analysis**. Overrepresented **Gene Ontology (GO)** terms were identified to elucidate the molecular pathways and immunological signatures associated with COVID-19 and bacterial pneumonia relative to healthy controls.

4. Results

4.1 Data acquisition

```
data_path <- here("GSE161731")

#Expression matrix
counts <- read.csv(here("GSE161731", "GSE161731_counts.csv.gz"),
                   row.names = 1, check.names = FALSE)

#with row.name = 1 we associate all the row data to the gene (which is de
tailed in the first column)

#Metadata
metadata <- read.csv(here("GSE161731", "GSE161731_key.csv.gz"),
                     header = TRUE, row.names = 1, check.names = FALSE) #
same argument, but in this case we are associating sample id (patient) wi
th the remaining data

#Checkpoint for data acquisition
length(intersect(colnames(counts), rownames(metadata)))

## [1] 195
```

We continue with the addition of chromosomic location:

```
# Removal of different gene versions
rownames(counts_clean) <- gsub("\\..*$", "", rownames(counts_clean))

#Annotation calling
edb <- EnsDb.Hsapiens.v86
```

```r
genes_coords <- genes(edb)

common_genes <- intersect(rownames(counts_clean), names(genes_coords))
length(common_genes)

## [1] 57602

counts_final <- counts_clean[common_genes, ]
coords_final <- genes_coords[common_genes]
metadata_final <- metadata_clean
```

Now, we can create the object: **summarizedExperiment**

```r
se <- SummarizedExperiment(
  assays = list(counts = counts_final),
  colData = DataFrame(metadata_final),
  rowRanges = coords_final
)

#We include metadata information (empty by defect)
metadata(se) <- list(
  GEO_accession = "GSE161731",
  organism = "Homo sapiens",
  assay_type = "RNA-seq"
)

#Definitive object: summarizedExperiment
se

## class: RangedSummarizedExperiment
## dim: 57602 195
## metadata(3): GEO_accession organism assay_type
## assays(1): counts
## rownames(57602): ENSG00000223972 ENSG00000227232 ... ENSG00000277475
##   ENSG00000268674
## rowData names(6): gene_id gene_name ... symbol entrezid
## colnames(195): 94189 DU09-03S0000604 ... 105847 105848
## colData names(8): subject_id age ... hospitalized batch
```

4.2 RNA-seq Data Preprocessing and Read Count Normalization

Now, we have 3 clean cohorts which contains 19 healthy individuals, 23 who suffered from bacterial pneumonia and 77 patients from COVID-19.

Once the data has been correctly preprocessed, we must **remove all the genes with low expression (probable biological noise)**.

```r
# Removal of genes with <10 total counts
keep <- rowSums(assay(se_clean, "counts")) >= 10

#Removal of these genes in which the 90% of the values correspond to zero
```

```
keep <- keep & rowSums(assay(se_clean, "counts") == 0) <= 0.9 * ncol(se_c
lean)

se_filtered <- se_clean[keep, ] #We maintain the aforementioned files (TR
UE under keep variable and all the columns)
print(paste("Genes after data cleaning processing:", sum(keep), "Final ge
nes for the study:", nrow(se_filtered)))

## [1] "Genes after data cleaning processing: 28258 Final genes for the s
tudy: 28258"
```

At the beginning we had near 60K genes. Now, we will work only with near to 30K (**50% of the genes are noise to the study**).

4.3 TMM and log Normalization

Following author's criteria, we will employ TMM (Trimmed Mean of M-values) normalization (Robinson & Oshlack, 2010) . With this methodology, the most central values will be averaging and the extreme values will be dropped out. Normalized counts were then transformed to log_2CPM with pseudo-count (prior.count=1) with this criteria:

TMM normalization

1. **For each sample vs. reference** (calculated by edgeR):
   - Compute log_2(count$_{\text{sample}}$/count$_{\text{reference}}$) **per gene**
   - **Trim** extreme values (75% outliers removed)
   - Average the central 25% → **TMM factor**
2. **Normalize**: $counts_{\text{TMM}} = raw\_counts/TMM\_factor$
3. **log2CPM**: $logCPM = log_2\left(\frac{counts_{\text{TMM}} \times 10^6}{library\_size} + 1\right)$

This double-normalization corrects both **composition bias** (TMM) and **library size** (CPM), stabilizing variance for downstream PCA/clustering.

TMM normalization: For each sample vs. reference sample (edgeR) - TMM factor considerate the number of reads of each gene - Calculate $log2(rawcounts/TMMfactor])$ for each single sample - Remove extreme values: atypical data - Average 25% of central values

$$normalized_counts = rawcounts/TMMfactor$$

$$logCPM = log2\big((normalized_counts * 1e6)/1e6 + prior.count\big)$$

```
library(edgeR)

# Extraction of raw data (counts) to create a new object: DEGList
```

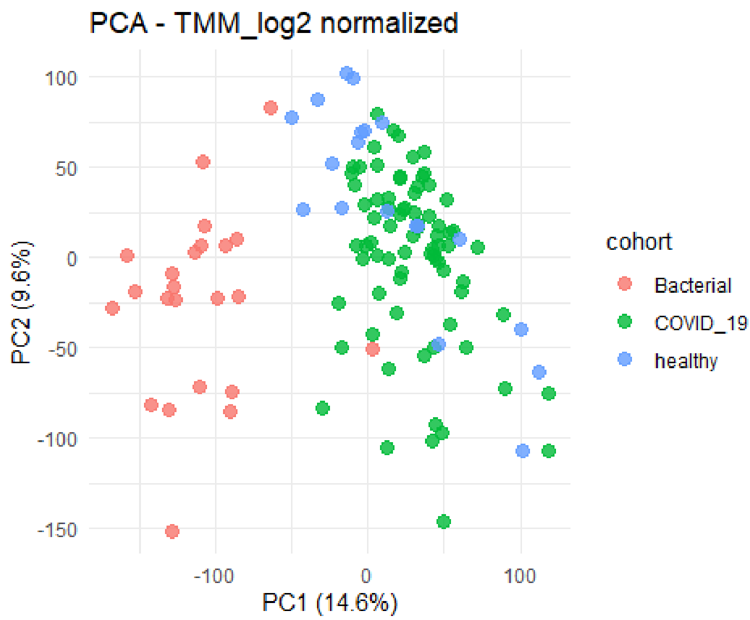
```
counts_matrix <- assay(se_filtered, "counts")
dge <- DGEList(counts = counts_matrix)
dge <- calcNormFactors(dge, method = "TMM") #TMM is already included in t
his package and ready to use

# Logarithmic transformation
logCPM <- cpm(dge, log = TRUE, prior.count = 1)

# We save this information as new assay
assay(se_filtered, "TMM_log2") <- logCPM
assayNames(se_filtered) #We have raw data and post-TMM data

## [1] "counts"   "TMM_log2"
```

4.4 Exploratory data analysis (EDA)

4.4.1 Visualization tools

4.4.1.1 PCA (Principal Component Analysis)

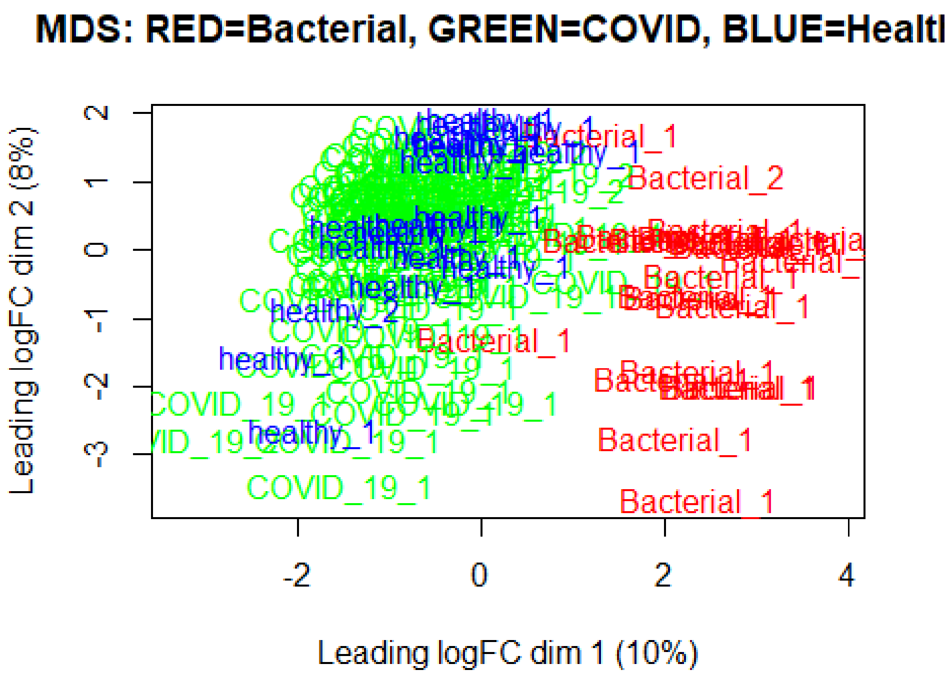
```
# Data extraction from TMM_log2
plot_data <- assay(se_filtered, "TMM_log2")
pca_res <- prcomp(t(plot_data), scale. = TRUE)

# Data frame for ggplot
pca_df <- data.frame(
  PC1 = pca_res$x[,1],
  PC2 = pca_res$x[,2],
  cohort = colData(se_filtered)$cohort,
  batch = colData(se_filtered)$batch,
  gender = colData(se_filtered)$gender,
  sample = colnames(se_filtered)
)

# PCA
ggplot(pca_df, aes(PC1, PC2, color = cohort)) +
  geom_point(size = 3, alpha = 0.8) +
  labs(title = "PCA - TMM_log2 normalized",
       x = paste0("PC1 (", round(summary(pca_res)$importance[2,1]*100,1),
"%)"),
       y = paste0("PC2 (", round(summary(pca_res)$importance[2,2]*100,1),
"%)")) +
  theme_minimal()
```



Visual analysis of the gene expression profiles reveals a **clear segregation of the bacterial infection samples**. In contrast, considerable **overlap is observed between the transcriptomes of COVID-19 patients and healthy controls**.

4.4.1.2 MDS (Multidimensional Scaling)

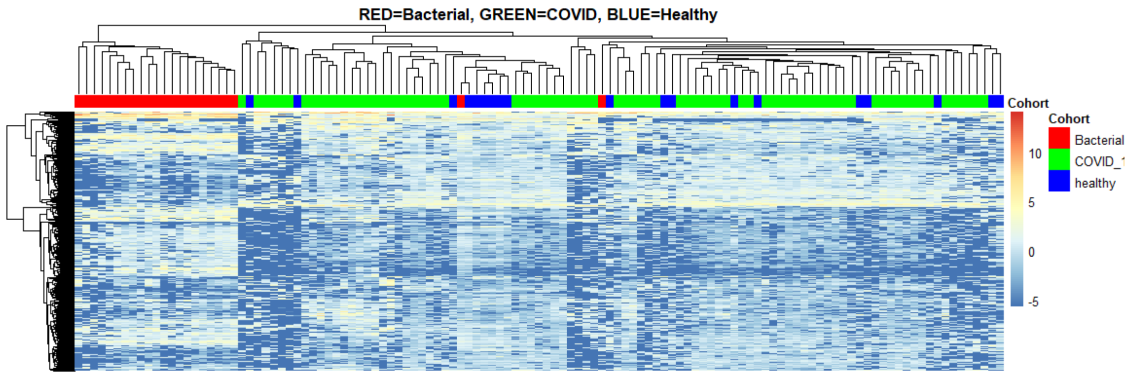
```
# MDS
palette <- c("Bacterial" = "red", "COVID_19" = "green", "healthy" = "blue
")
cohort_factor <- factor(colData(se_filtered)$cohort)
palette_mds <- palette[cohort_factor]

plotMDS(dge,
        col = palette_mds,
        labels = paste(colData(se_filtered)$cohort,
                       colData(se_filtered)$batch, sep="_"),
        main = "MDS: RED=Bacterial, GREEN=COVID, BLUE=Healthy")
```



Both analyses (PCA and MDS) demonstrate a **consistent sample segregation**, confirming the robustness of the overall clustering pattern.

4.4.1.3 Heatmap

```
tmm_data <- assay(se_filtered, "TMM_log2")
tmm_clean <- tmm_data; tmm_clean[is.infinite(tmm_data) | is.na(tmm_data)]
<- 0

# Top 1000 genes variables
rv <- apply(tmm_clean, 1, var, na.rm = TRUE); rv[is.na(rv)] <- 0
top_genes <- order(rv, decreasing = TRUE)[1:1000]

annotation_col <- data.frame(Cohort = colData(se_filtered)$cohort)
rownames(annotation_col) <- colnames(tmm_clean)
annotation_colors <- list(Cohort = c("Bacterial" = "red", "COVID_19" = "g
reen", "healthy" = "blue"))
pheatmap(tmm_clean[top_genes, ],
         annotation_col = annotation_col,
         annotation_colors = annotation_colors,
         show_rownames = FALSE,
         show_colnames = FALSE,
         main = "RED=Bacterial, GREEN=COVID, BLUE=Healthy")
```

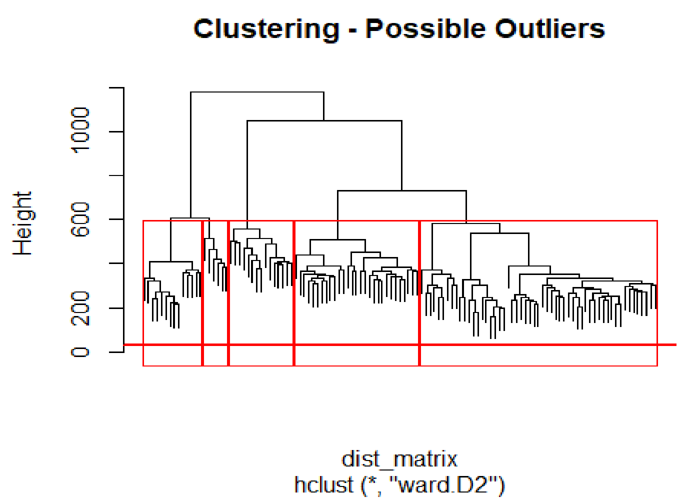


The hierarchical clustering dendrogram reveals **distinct separation of the Bacterial cohort** (red annotation bar), while **COVID-19 and healthy samples show partial overlap**. This confirms previous visualization tools, such as PCA.

4.4.2 Atypical data identification

We select **Ward method to calculate minimum variance inside each cluster** (version D2).

```r
# Distance calculation
dist_matrix <- dist(t(tmm_clean))
hclust_res <- hclust(dist_matrix, method = "ward.D2")

# Plot to visualize outliers
plot(hclust_res, labels = FALSE, main = "Clustering - Possible Outliers")
abline(h = 30, col = "red", lwd = 2)  # Cutoff
rect.hclust(hclust_res, k = 5, border = "red")
```

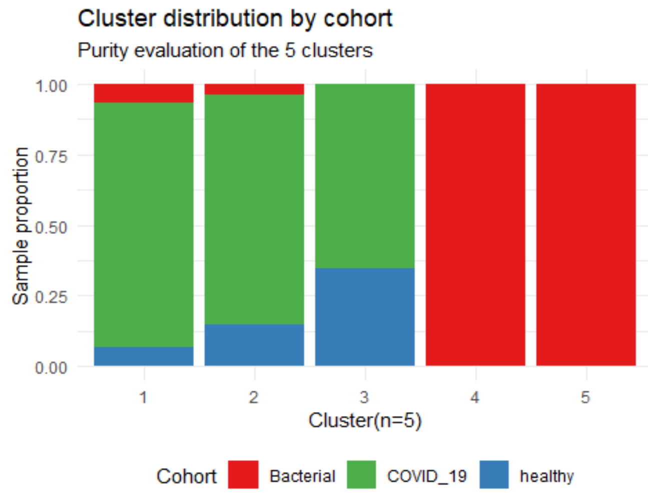


With this method, 5 different clusters were created. Now, we must see the possible correspondence of these clusters with biological cohorts.

```r
# We define the previous clusters
clusters_5 <- cutree(hclust_res, k = 5)

# We extract cohort data from summarizedExperiment
comparison_df <- data.frame(
  Sample = names(clusters_5),
  Cluster = as.factor(clusters_5),
  Cohort = colData(se_filtered)$cohort
)

# Table
contingency_table <- table(Cluster = comparison_df$Cluster, Cohort = comp
arison_df$Cohort)
print(contingency_table)

##         Cohort
## Cluster Bacterial COVID_19 healthy
##       1         1       13       1
##       2         2       45       8
##       3         0       19      10
##       4         6        0       0
##       5        14        0       0
```

This contingency table shows a clear overlap in COVID-19 and control patients in cluster 2. However, bacterial pneumonia patients are clearly separated in clusters 4 and 5.

```r
ggplot(comparison_df, aes(x = Cluster, fill = Cohort)) +
  geom_bar(position = "fill") + # "fill" muestra proporciones (0 a 1)
  theme_minimal() +
  scale_fill_manual(values = c("Bacterial" = "#E41A1C", "COVID_19" = "#4D
AF4A", "healthy" = "#377EB8")) +
  labs(title = "Cluster distribution by cohort",
       subtitle = "Purity evaluation of the 5 clusters",
       x = "Cluster(n=5)",
       y = "Sample proportion") +
  theme(legend.position = "bottom")
```



Now, we check if tiny clusters (with 2 individuals) exist. These patients will be considered as outliers.

```
# Identification of small clusters (n <= 2)
counts_por_cluster <- table(clusters_5)
clusters_outliers <- names(counts_por_cluster[counts_por_cluster <= 2])

# Extraction of the samples within this criteria
final_outliers <- names(clusters_5[clusters_5 %in% clusters_outliers])

# Cleaning up
se_clean <- se_filtered[, !(colnames(se_filtered) %in% final_outliers)]
```

After this short verification, we will continue the identification of atypical data. Based on Spearman correlation matrix for each individual. The patients with -2SD or +2SD with respect to the mean will be considered as outlier:

```
sample_cor <- cor(tmm_clean, method = "spearman")
mean_cor <- colMeans(sample_cor)

# Threshold
threshold_cor <- mean(mean_cor) - 2 * sd(mean_cor)
outliers_num <- names(mean_cor[mean_cor < threshold_cor])

#Outliers removal
se_clean <- se_filtered[, !(colnames(se_filtered) %in% outliers_num)]
message(paste("Analysis contains", ncol(se_clean), "samples."))
```

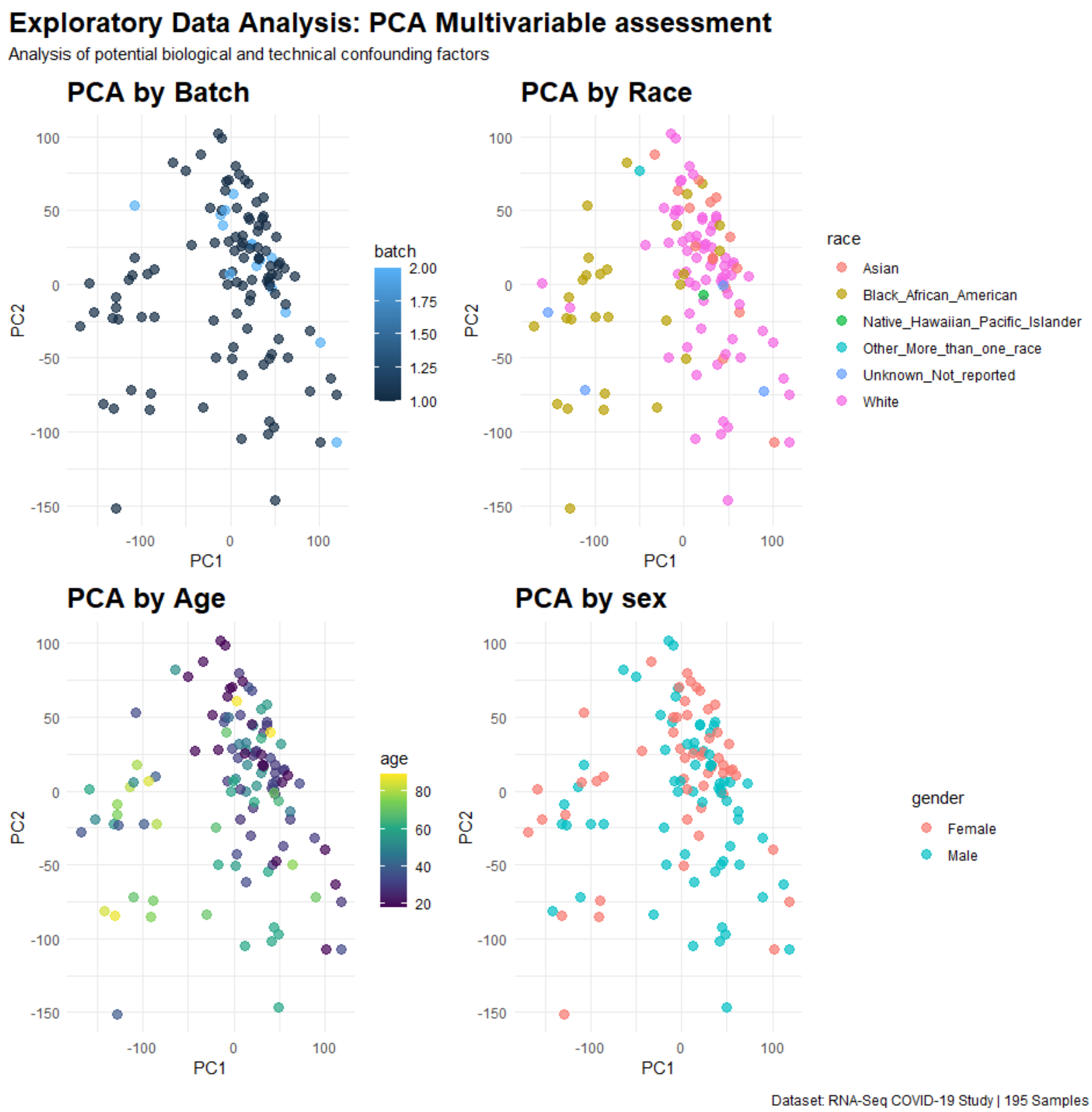
We continue with 112 samples.

4.4.3 Confounding variables

We will visualize PCA by different confounding variables: unmeasured third variables that influences both the supposed cause and the supposed effect, potentially creating a false mathematical correlation between them.

```
# Variables to consider
pca_df$batch        <- colData(se_filtered)$batch
pca_df$age <- as.numeric(colData(se_filtered)$age)
pca_df$race <- colData(se_filtered)$race
pca_df$gender <- colData(se_filtered)$gender
pca_df$hospitalized <- colData(se_filtered)$hospitalized
pca_df$onset        <- colData(se_filtered)$time_since_onset

# All the variables employed must be factors
# Healthy controls must be the reference cohort
se_filtered$cohort <- factor(se_filtered$cohort)
se_filtered$cohort <- relevel(se_filtered$cohort, ref = "healthy")

se_filtered$batch  <- factor(se_filtered$batch)
se_filtered$race   <- factor(se_filtered$race)
se_filtered$age <- scale(as.numeric(as.character(se_filtered$age)))

#We have 42 patients with no data for variables: hospitalized and onset.
Probably, they are healthy controls
table(pca_df$cohort, is.na(pca_df$hospitalized))

##
##             FALSE TRUE
##   Bacterial     0   23
##   COVID_19     77    0
##   healthy       0   19
```

All patients were hospitalized, whereas none of the healthy controls were. Consequently, **hospitalization status is a nested variable and should not be included as a confounding factor** in the analysis of the full cohort.



Furthermore, visual inspection reveals that Batch 2 is predominantly localized to the right side of the plot, while Batch 1 exhibits a broader distribution. **Including batch as a covariate in the model is highly recommended**, as it is a standard approach to mitigate potential technical artifacts arising from the sequencing workflow Hicks et al. (2018).

Regarding **race**, White and African American individuals constitute the vast majority of the cohort. There appears to be a **clear distinction between these two racial groups**.

There is **no clear separation between males and females**; thus, biological sex was not considered a confounding variable. In contrast, noticeable differences are observed regarding the age of the study subjects. Consequently, **age was included as a covariate in the model**.

4.6. Differential Gene Expression (DGE) Analysis: Functional Enrichment and Interpretation

We will implement a **generalized linear model (GLM)** incorporating race, age, and batch as confounding covariates to isolate their effects from the cohort variable. Our selected method is **DESeq2**.

The primary objective of creating the DESeqDataSet object is to **evaluate differential gene expression across cohorts while ensuring the mitigation of technical and biological noise** caused by these confounding variables.

```
register(SnowParam(workers = 6))
# Matrix construction
dds <- DESeqDataSet(se_filtered, design = ~ batch + race + age + cohort)

# Execution of the analysis (dispersion estimation and model fitting
# Shrinkage is included in this step
dds <- DESeq(dds, parallel= T)
```

Next, we specify the contrast vectors to compare COVID-19 vs. healthy and Bacterial pneumonia vs. healthy, allowing us to extract the differential expression results for each condition.

```
# Contrast: COVID-19 vs Healthy
res_covid <- results(dds,
                     contrast = c("cohort", "COVID_19", "healthy"),
                     alpha = 0.05) # Umbral de significación

# Contrast: Bacterial vs Healthy
res_bacterial <- results(dds,
                         contrast = c("cohort", "Bacterial", "healthy"),
                         alpha = 0.05)

# Summaries
message("DEGs: COVID-19 vs Healthy")
summary(res_covid)

##
## out of 28258 with nonzero total read count
## adjusted p-value < 0.05
## LFC > 0 (up)       : 545, 1.9%
## LFC < 0 (down)     : 1366, 4.8%
## outliers [1]       : 338, 1.2%
## low counts [2]     : 12826, 45%
## (mean count < 35)
## [1] see 'cooksCutoff' argument of ?results
## [2] see 'independentFiltering' argument of ?results
```

```r
message("DEGs: Bacterial vs Healthy")
summary(res_bacterial)

##
## out of 28258 with nonzero total read count
## adjusted p-value < 0.05
## LFC > 0 (up)       : 3013, 11%
## LFC < 0 (down)     : 4221, 15%
## outliers [1]       : 338, 1.2%
## low counts [2]     : 6286, 22%
## (mean count < 4)
## [1] see 'cooksCutoff' argument of ?results
## [2] see 'independentFiltering' argument of ?results
```

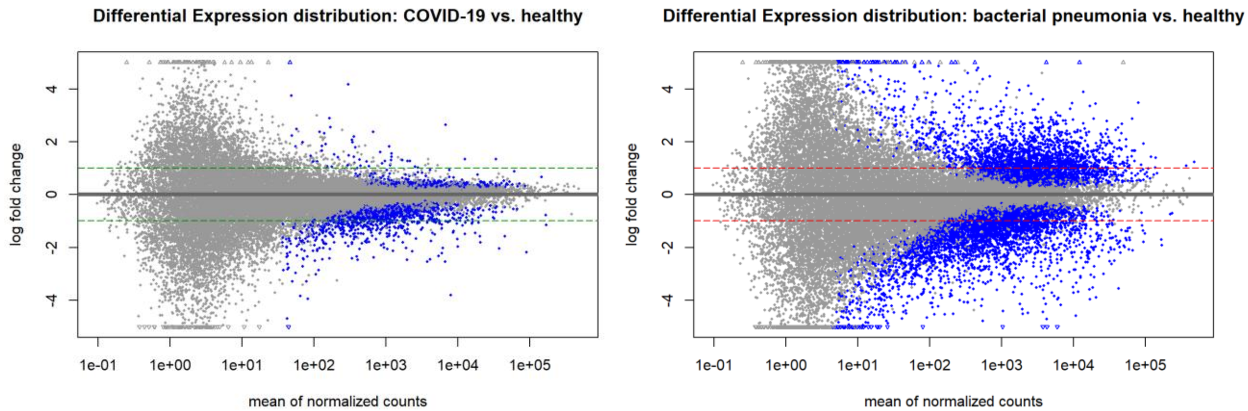
Differential gene expression analysis using the **negative binomial distribution (DESeq2) with Benjamini-Hochberg correction by FDR** revealed significant transcriptomic alterations in both infection cohorts compared to the healthy control group.

The **bacterial cohort exhibited a more extensive transcriptional response, with approximately 26% of genes showing differential expression (**$padj < 0.05$**)**, compared to **6.8% in the COVID-19 group**. Notably, the COVID-19 analysis required a more stringent independent filtering, with 47% of genes excluded due to low mean counts.

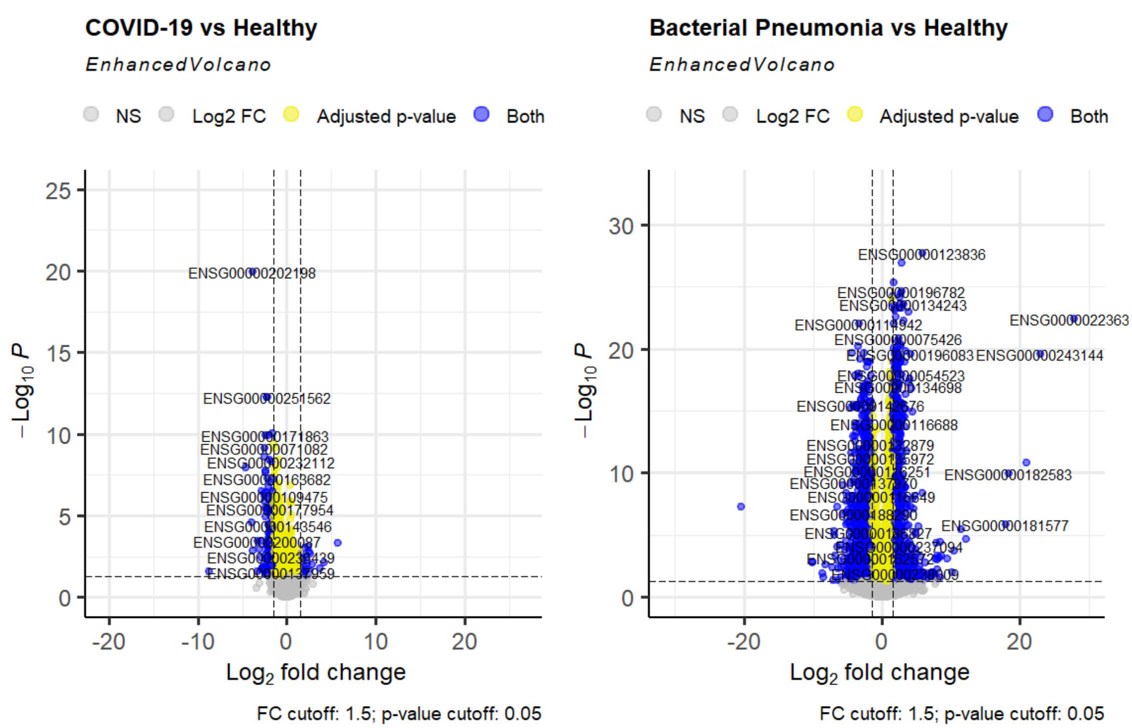
Outlier detection remained consistent across both comparisons, accounting for approximately 2% of the total features. These results highlight a distinct molecular signature for each pathogen, with bacterial pneumonia triggering a broader systemic gene dysregulation than COVID-19 in this dataset.

4.6.1 Visualization of DEG (Differential Expressed Genes)

```r
plotMA(res_covid, ylim=c(-5, 5),
       main="Differential Expression distribution: COVID-19 vs. healthy")
# Reference in log2FoldChange = 1
abline(h=c(-1, 1), col="green4", lty=5)

plotMA(res_bacterial, ylim=c(-5, 5),
       main="Differential Expression distribution: bacterial pneumonia vs
. healthy")
# Reference in log2FoldChange = 1
abline(h=c(-1, 1), col= "red2", lty=5)
```

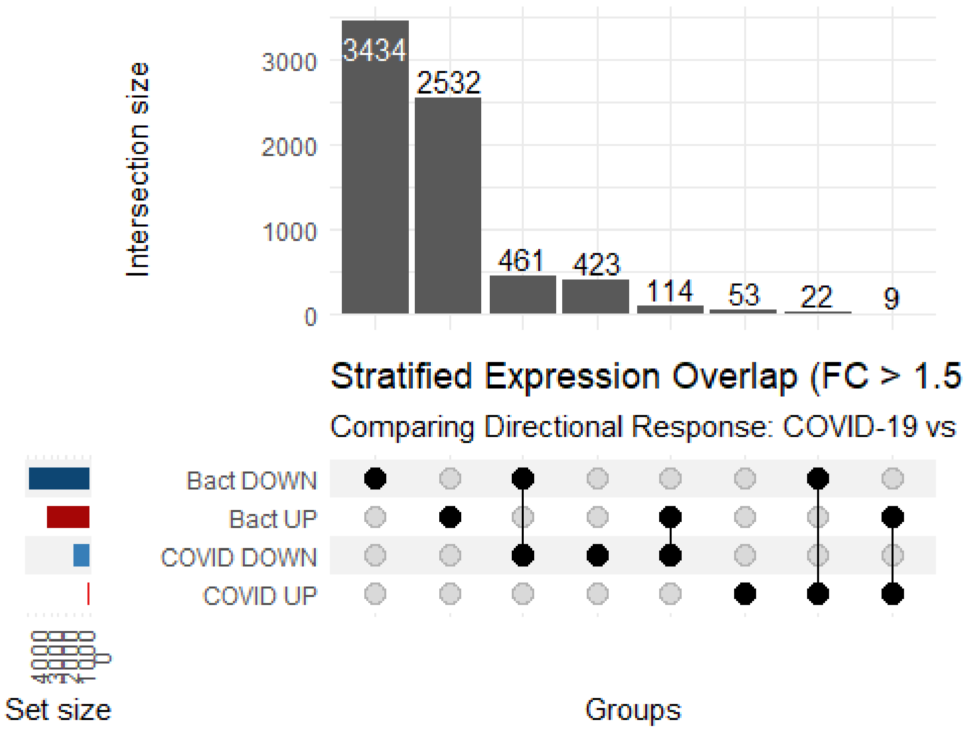


The **comparative differential expression analysis reveals distinct transcriptional landscapes between bacterial and viral etiologies**. While bacterial pneumonia induces a massive and symmetric inflammatory response characterized by a high number of significantly dysregulated genes, the COVID-19 signature appears more targeted. Notably, the COVID-19 profile exhibits a prominent cluster of down-regulated genes among high-abundance transcripts, suggesting a specific suppressive effect on host cellular processes.



To gain an overview of coincidences between the two cohorts we generate an upset plot. We use a threshold of foldchange 1.5 and alpha = 0.05 to represent Differential Expressed Genes (DEGs).

```r
# Threshold definition
lfc_threshold <- log2(1.5)

# Gene list
list_input <- list(
  `COVID UP`   = rownames(res_covid[which(res_covid$padj < 0.05 & res_cov
id$log2FoldChange > lfc_threshold), ]),
  `COVID DOWN` = rownames(res_covid[which(res_covid$padj < 0.05 & res_cov
id$log2FoldChange < -lfc_threshold), ]),
  `Bact UP`    = rownames(res_bacterial[which(res_bacterial$padj < 0.05 &
res_bacterial$log2FoldChange > lfc_threshold), ]),
  `Bact DOWN`  = rownames(res_bacterial[which(res_bacterial$padj < 0.05 &
res_bacterial$log2FoldChange < -lfc_threshold), ])
)

# Binary vector
all_genes <- unique(unlist(list_input))
upset_df <- data.frame(gene = all_genes)
for(set_name in names(list_input)) {
  upset_df[[set_name]] <- as.integer(upset_df$gene %in% list_input[[set_n
ame]])
}

# colNames define the genes
sets_indices <- names(list_input)

#Upset plot
upset(
  upset_df,
  sets_indices, # Vector with colNames
  name = "Groups",
  width_ratio = 0.1,
  queries = list(
    upset_query(set = "COVID UP", fill = "#E41A1C"),
    upset_query(set = "Bact UP", fill = "#A60606"),
    upset_query(set = "COVID DOWN", fill = "#377EB8"),
    upset_query(set = "Bact DOWN", fill = "#0D4673")
  ),
  base_annotations = list(
    'Intersection size' = intersection_size(counts = TRUE)
  ),
  set_sizes = upset_set_size() + theme(axis.text.x = element_text(angle =
90))
) + labs(
    title = "Stratified Expression Overlap (FC > 1.5)",
    subtitle = "Comparing Directional Response: COVID-19 vs Bacterial Pne
umonia"
)
```



The **intersection analysis confirms a more extensive transcriptional landscape in Bacterial pneumonia compared to COVID-19** (both in up-regulated and down-regulated genes) relative to healthy controls. Furthermore, the intersections reveal that **overlaps between both datasets occur primarily in a concordant manner**: genes up-regulated in COVID-19 tend to be up-regulated in the bacterial cohort, with a similar pattern observed for down-regulated genes.

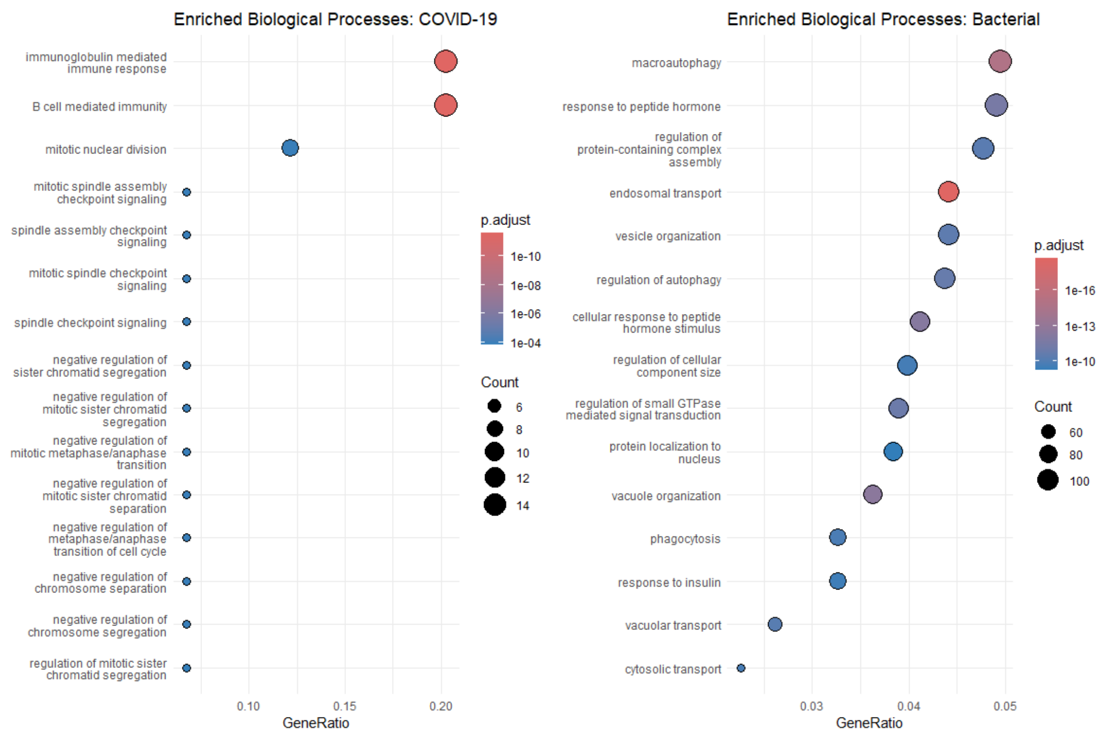
These results identify a common molecular response to respiratory infections, highlighting **potential biomarkers for general lung inflammation and host defense mechanism**.

4.7 Functional enrichment analysis

We select the upregulated genes from COVID-19 and pneumonia patients (over healthy controls) to study the main mechanisms and physiological pathways activated under these conditions.

```
covid_up <- list_input[["COVID UP"]]
bact_up  <- list_input[["Bact UP"]]
covid_up_clean <- as.character(covid_up)
bact_up_clean  <- as.character(bact_up)

enr_covid <- enrichGO(gene          = covid_up_clean,
                      OrgDb         = org.Hs.eg.db,
                      keyType       = 'ENSEMBL',
                      ont           = "BP",
```

```
                      pAdjustMethod = "BH",
                      pvalueCutoff  = 0.05,
                      qvalueCutoff  = 0.2)

enr_bact <- enrichGO(gene          = bact_up_clean,
                     OrgDb         = org.Hs.eg.db,
                     keyType       = 'ENSEMBL',
                     ont           = "BP",
                     pAdjustMethod = "BH",
                     pvalueCutoff  = 0.05,
                     qvalueCutoff  = 0.2)

#Visualization of the most enriched biological process in dotplot
enr_covid <- setReadable(enr_covid, OrgDb = org.Hs.eg.db, keyType="ENSEMB
L") #Transformation of gene nomenclature
enr_bact  <- setReadable(enr_bact, OrgDb = org.Hs.eg.db, keyType="ENSEMBL
")

# Plots
dot_covid <- dotplot(enr_covid, showCategory=15) +
  ggtitle("Enriched Biological Processes: COVID-19") +
  theme_minimal()

dot_bact <- dotplot(enr_bact, showCategory=15) +
  ggtitle("Enriched Biological Processes: Bacterial") +
  theme_minimal()

dot_covid | dot_bact
```



Functional enrichment analysis reveals deep contrast between the two infectious etiologies despite their shared clinical presentation. In COVID-19, the host response is predominantly characterized by immediate immune activation, specifically through immunoglobulin-mediated responses and B-cell-mediated cellular immunity. To a lesser extent, pathways related to cell division are also enriched, potentially reflecting lymphocyte proliferation.

In contrast, the bacterial infection signature is directly linked to **cellular structural and degradative processes, specifically autophagy, vesicle-mediated transport, and endocytosis**. These pathways likely represent the host's attempt to internalize and degrade bacteria.

5. Discussion

The role of **Bioconductor in gene expression analysis** is invaluable for modern biomedical research. Currently, scientific standards require researchers to **deposit high-throughput sequencing data in public repositories such as the GEO (Gene Expression Omnibus)** Database. These datasets are typically available in both raw and processed formats, providing a transparent foundation for the scientific community.

For any bioinformatician the ability to interpret and re-analyze these studies is an essential skill. Public data serves as a powerful tool for comparative and complementary analysis, allowing for the validation of in-house findings against established benchmarks. Furthermore, leveraging the current state-of-the-art through existing experiments significantly optimizes research resources, reducing the need for redundant laboratory testing and accelerating the discovery of disease-specific molecular signatures.

In this study, among the top 15 enriched biological processes in COVID-19 and bacterial pneumonia, there is **minimal overlap between the two conditions**. This suggests that the molecular mechanisms driving the host's defense are highly pathogen-specific, offering a clear transcriptomic basis for differential diagnosis.

6. Reproducibility statement

All raw transcriptomic data and corresponding clinical metadata analyzed in this study are publicly available through the NCBI Gene Expression Omnibus (GEO) under the accession number **GSE161731**. This ensures that the foundational evidence for our findings remains accessible for independent verification.

The complete analytical pipeline, including the R Markdown (.Rmd) source code, pre-processing scripts, and high-resolution figures, is hosted on **GitHub** (https://github.com/lnungon/RNASeq-Differential-Expression-SARS-CoV-2). The repository is structured to allow for full computational replication of the results, from raw count processing to functional enrichment.

To ensure computational reproducibility, detailed information regarding the R environment, including package versions and system dependencies, is documented in the accompanying *session_info.txt* file.

7. Bibliography

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